

SURAJ PATIL

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PROFESSIONAL SUMMARY

Final-year Computer Engineering student with strong hands-on experience in **Full Stack Web Development** (React.js, Node.js, Express.js, MongoDB) and **Machine Learning** (Python, Scikit-learn, Pandas). Proven track record of building real-time applications, secure backend systems with ACID transactions, automated vulnerability monitoring pipelines, and predictive ML models. Consistently high academic achiever (**9.22 TE CGPA**) seeking Software Engineering opportunities.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, C++
Frontend: React 19, HTML5, CSS3, Tailwind CSS, DaisyUI, Zustand, Vite, Axios
Backend: Node.js, Express.js, RESTful APIs, Socket.IO (WebSockets), JWT Authentication, OAuth2
Databases: MongoDB, Mongoose (ACID Transactions), MySQL
AI / ML: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn, Supervised & Unsupervised ML, K-Means Clustering
Tools & Services: Git, GitHub, Postman, VS Code, Selenium, OpenPyXL, MongoDB Atlas, AWS Fundamentals, Cloudinary, Resend, Arcjet

PROFESSIONAL EXPERIENCE

Web Development Intern — *The Developers Arena*

Jan 2026 – Mar 2026

- Developed modular, reusable React UI components and responsive layouts for an E-Commerce web application.
- Integrated RESTful API endpoints for seamless dynamic client-side rendering and active state updates.
- Contributed to frontend debugging, cross-device layout optimization, and quality testing across modern browsers.
- Participated in active Git version control workflows, collaborative code reviews, and software engineering practices.

KEY PROJECTS

QuickChat — Real-Time Chat Platform | [React 19](#), [Node.js](#), [Express](#), [MongoDB](#), [Socket.IO](#), [Zustand](#), [Arcjet](#)

- Engineered a full-stack real-time messaging application supporting instant messaging, typing indicators, user presence, read receipts, and message editing/deletion.
- Implemented WebSockets via Socket.IO alongside Express REST routes, managing UI state efficiently with Zustand.
- Secured authentication using JWTs in HttpOnly cookies, integrated Cloudinary for image sharing, and applied Arcjet rate limiting.

Bank Transaction System | [Node.js](#), [Express.js](#), [MongoDB](#), [Mongoose](#), [JWT](#), [Nodemailer](#)

- Architected a secure backend banking system featuring user authentication, account management, fund transfers, and token blacklist logout.
- Utilized MongoDB session-based **ACID transactions** to guarantee data consistency and atomic rollbacks during transfer failures.

- Implemented an immutable ledger-based balance calculation model and idempotency key handling to prevent double-spending.

VulnTracker — Automated OEM Vulnerability Reporting System | [Python](#), [Selenium](#), [React](#), [MongoDB](#), [OpenPyXL](#)

- Developed an automated security monitoring system collecting CVE advisories from Microsoft, Cisco, and Google security portals.
- Built Selenium web scrapers with regex extraction to filter Critical/High advisories, storing metrics in MongoDB.
- Created a React admin dashboard alongside automated OpenPyXL Excel report generation and email alert notifications.

CreditWise Loan Approval System & SmartCart Customer Segmentation | [Python](#), [Scikit-learn](#), [Pandas](#), [NumPy](#), [K-Means](#)

- Trained supervised classification models (Logistic Regression, Random Forest) evaluated via Precision, Recall, and ROC-AUC for credit risk prediction.
- Applied unsupervised K-Means clustering on RFM transaction data, optimizing clusters via Elbow Curves and Silhouette Scores.

EDUCATION

Bachelor of Engineering in Computer Engineering

2023 – 2027

Marathwada Mitra Mandal's College of Engineering (MMCOE), Pune — SPPU

- **Academic Performance:** TE CGPA: **9.22** | SE CGPA: **9.16** | FE CGPA: **9.82** | Sem 6 SGPA: **9.43** | Sem 5 SGPA: **9.00**

CERTIFICATIONS & HONORS

- **Certifications:** AWS Cloud Fundamentals (AWS, 2026), Data Science for Engineers (NPTEL, 2026), Database Management Systems (NPTEL, 2025), C++ Programming (Udemy, 2025).
- **Achievements:** 1st Prize Winner — Dexterity Technical Event (MMCOE) | Participant — OTIS Made to Move Community Mobility Programme.